

Customer Shopping Behavior Analysis

3,900 purchases, \$233,081 in revenue

Every customer spends about \$60, so volume drives revenue

Purchases

3,900

Total revenue

\$233K

Average purchase

\$59.76

Average rating

3.75

Group differences come from how many purchases each makes, not purchase size.

Total revenue \$233,081

From CSV to dashboard in four steps

1

Python (pandas)

Load, explore, clean the
3,900 rows and 18
columns

2

Fix and add

Fix 37 missing ratings with
category medians; add age
group and purchase
frequency in days

3

PostgreSQL

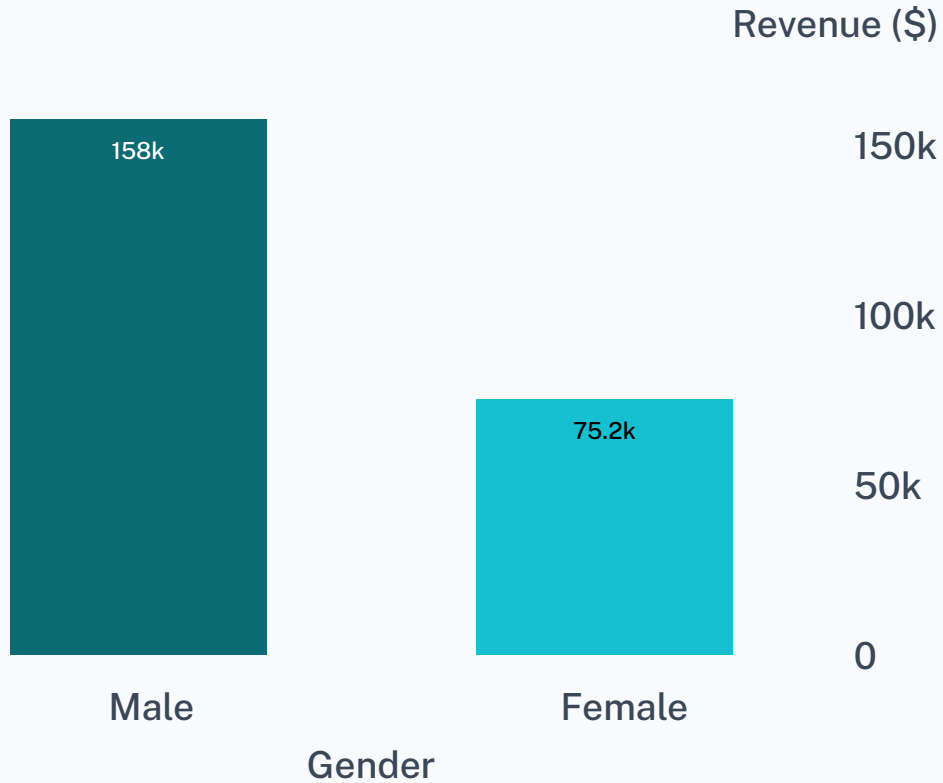
10 business queries using
CTEs and window
functions

4

Power BI

Dashboard with slicers for
non-technical users

Men bring 68% of revenue by making twice as many purchases



- **Male:** \$157,890 from 2,652 purchases, \$59.54 average
- **Female:** \$75,191 from 1,248 purchases, \$60.25 average
- Women spend slightly more per purchase but are only 32% of the base

Subscribers spend no more than non-subscribers

SUBSCRIBERS

\$59.49

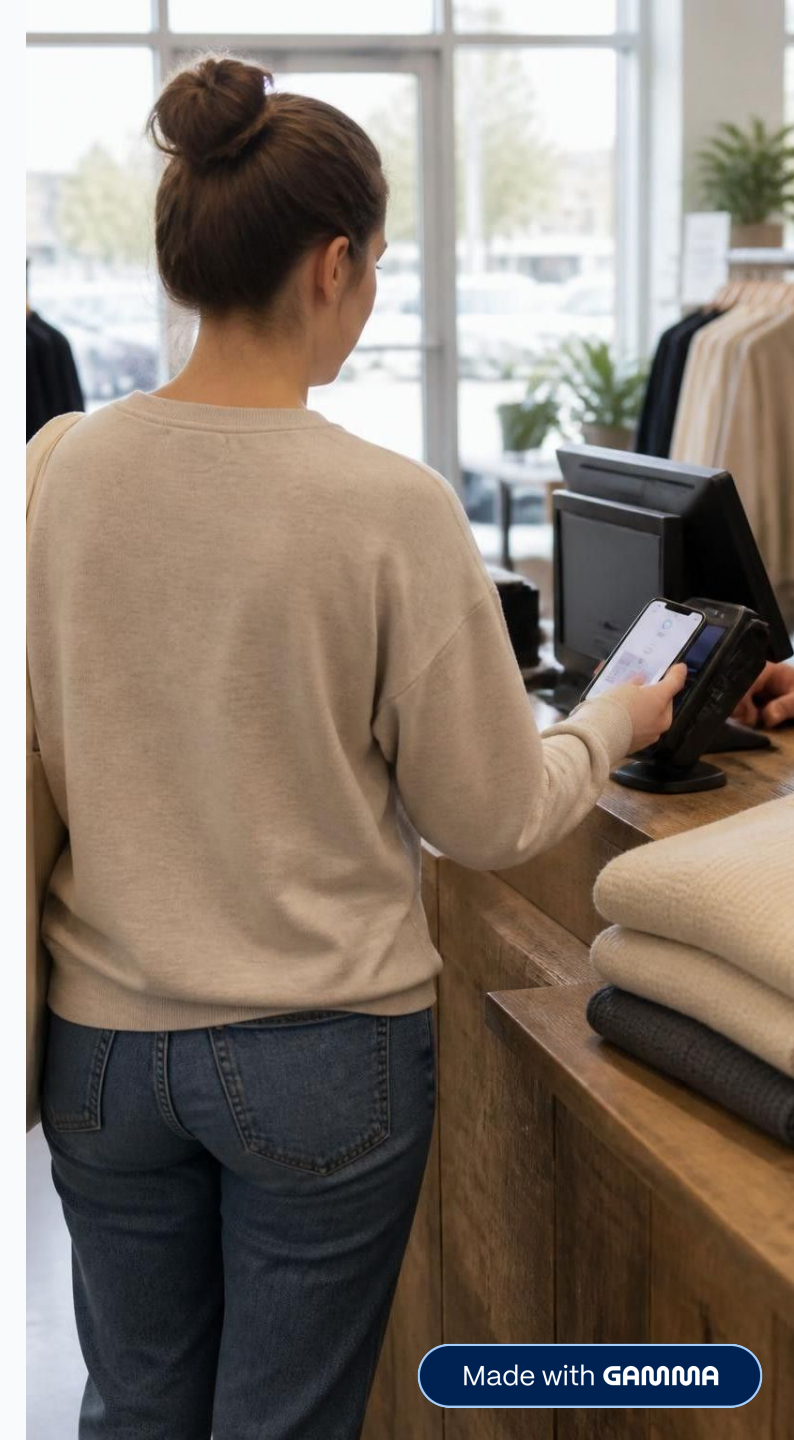
27% of purchases, \$62,645 revenue

NON-SUBSCRIBERS

\$59.87

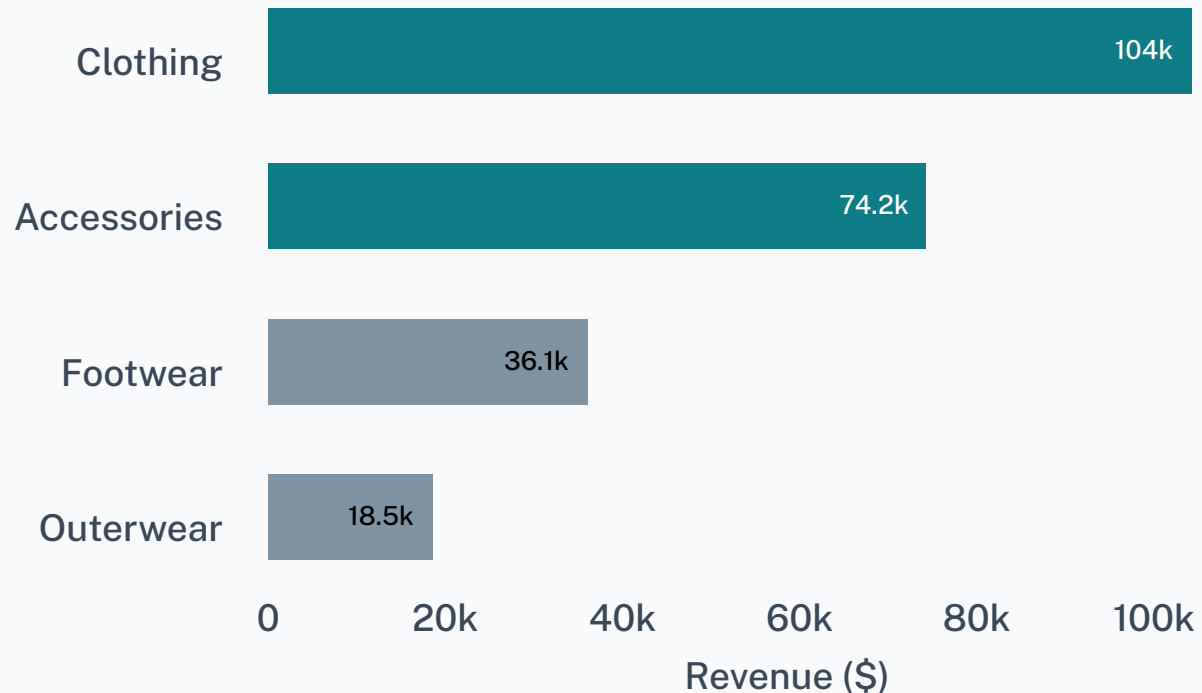
73% of purchases, \$170,436 revenue

Even among repeat buyers, 958 subscribe and 2,518 do not.



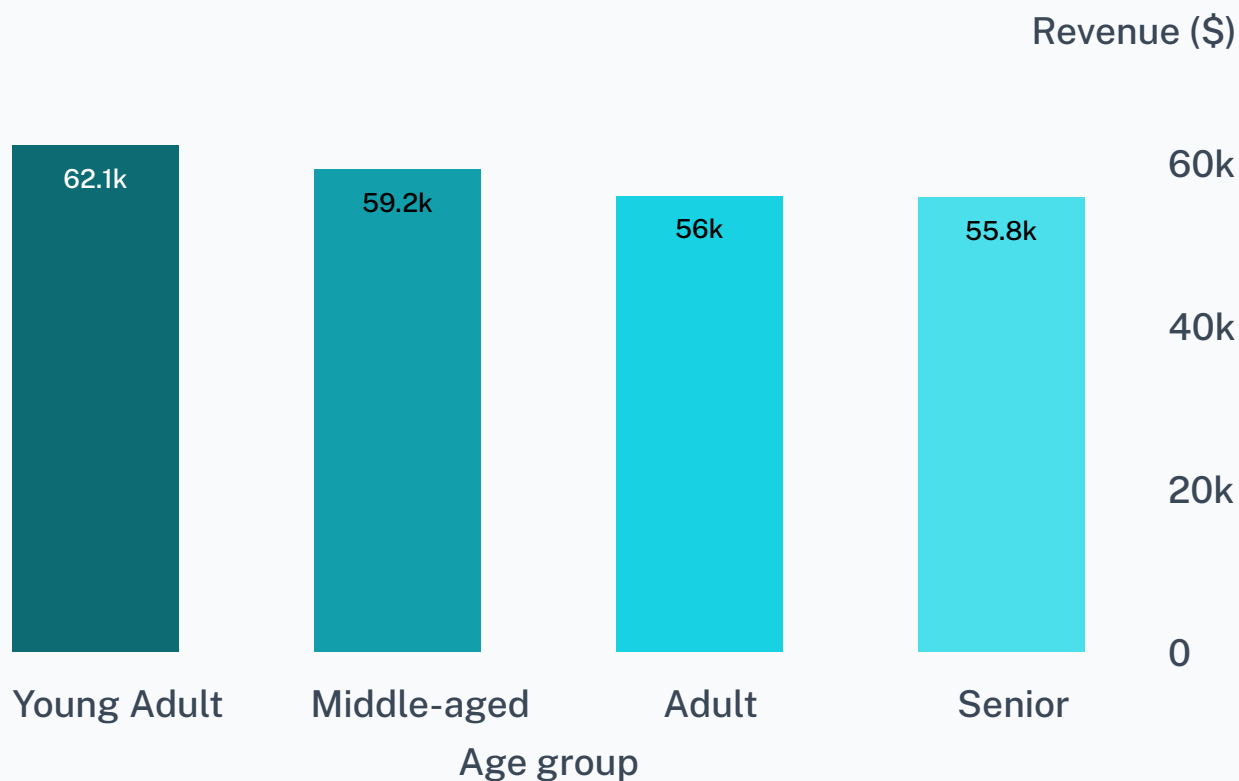
Clothing and Accessories earn 77% of revenue

Category



- Clothing \$104,264 (44.7%), Accessories \$74,200 (31.8%)
- Footwear \$36,093 (15.5%), Outerwear \$18,524 (7.9%)

Revenue is spread evenly across age groups



- Young Adult (18-31) leads at \$62,143 from 1,028 purchases
- Middle-aged \$59,197, Adult \$55,978, Senior \$55,763

Nearly half of Hat, Sneaker and Coat sales use a discount

43% of all purchases used a discount.

ITEM	DISCOUNT SHARE
Hat	50.0%
Sneakers	49.7%
Coat	49.1%
Sweater	48.2%
Pants	47.4%



Shipping type and ratings barely move the needle

AVERAGE PURCHASE BY SHIPPING

\$58.46

Standard

\$60.73

2-Day

TOP RATED

- Gloves 3.86
- Sandals 3.84
- Boots 3.82
- Hat 3.80
- Handbag 3.78

80% of customers are already loyal

80%

Loyal

Over 10 previous purchases: 3,116

18%

Returning

2-10 purchases: 701

2%

New

Only 83